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Seminar paper

Betting against beta

a study of proposition 1 & 2 in american equity markets

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Abstract

This paper examines the beta market anomaly, where stocks with higher betas tends to have lower alphas than those with lower betas, contradicting a fundamental prediction by the CAPM-model. This is done by testing the proposition 1 and 2 from the paper of Frazzini and Pedersen (2014). Using data from US equity markets we test proposition 1 by examining the relationship between beta and alpha to test for the presence of the market anomaly. To test proposition 2, we construct a BAB-portfolio to examine the performance of a strategy which exploits this market anomaly by going long on low beta stocks and short on high beta stocks. We find evidence of a negative relationship between beta and alpha, which supports proposition 1 and that the beta anomaly is present leading to a flatter market security line. We also find that the BAB-portfolio generates positive returns, which supports proposition 2. However, when we introduce transaction costs into our analysis, we find that the performance of the BAB-portfolio is significantly reduced, suggesting that the strategy may not be as effective as initially thought when accounting for real-world trading frictions.

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Chapter 1

Introduction

One of the central predictions of the capital asset pricing model (CAPM) is the linear relationship between expected returns and systematic risk which is undiversifiable. This relationship however has long been challenged by empirical evidence which suggests a much flatter line than predicted by the model. A reason for this, suggested by the literature such as Fama and French (2004) [1], is the presence of the market anomalies such as high-beta assets yielding lower returns compared to lower beta assets completely contradicting the model expectations. This is called the beta anomaly and a way of exploiting this anomaly is the investing strategy of betting against beta (BAB).

Introduced by Frazzini and Pedersen (2014) [2] the BAB-strategy is a simple long-short strategy which exploits the beta anomaly by going long on low beta stocks and short on high beta stocks. The strategy has been shown to generate positive returns when varying the markets and time-periods making it an attractive investment strategy for investors looking to capitalize on the beta anomaly. However, the performance of the BAB-strategy is not without controversy, with some studies suggesting that the strategy may not be as effective as initially thought, and that its performance may be driven by other factors such as size and value effects which is also documented by Frazzini and Pedersen (2014) [2].

To examine the relevance of the beta anomaly we test the propositions 1 and 2 from the paper of Frazzini and Pedersen (2014) [2] . The first proposition states that there is a negative relationship between beta and alpha, which implies that stocks with higher betas tend to have lower alphas than those with lower betas. The second proposition states that a strategy which exploits this market anomaly by going long on low beta stocks and short on high beta stocks, known as the BAB-portfolio, should generate positive returns.

For best understanding and comprehension of the subject matter the paper structure is as follows: first, we introduce the theoretical background and framework for the CAPM and why the existence of the beta anomaly is possible according to existing literature. Second, we describe our data and methodology for testing the anomaly and the performance of the BAB-strategy and how we introduce transaction costs as an extension of the analysis of Frazzini and Pedersen (2014) [2]. This is equivalent to testing proposition 1 and 2. Third, we present the results. Fourth, we connect our results to the existing literature and the implications. Lastly, we conclude and suggest avenues for future research.

Using US equity data for the period 1990-2020 we find a negative relationship between beta and alpha, which supports proposition 1 and that the beta anomaly is present leading to a flatter market security line. We also find that the BAB-portfolio generates positive returns, which supports proposition 2. However, when we introduce transaction costs into our analysis, we find that the performance of the BAB-portfolio is significantly reduced, suggesting that the strategy may not be as effective as initially thought when accounting for real-world trading frictions.

Chapter 2

Theory

In this chapter we discuss the theoretical background needed in order to understand the results. This entails firstly describing the CAPM as the baseline model of comparison and its predictions. Secondly, we discuss the shortcomings of the CAPM and how the presence of leverage constraints can lead to a flatter security market line thus creating the beta anomaly as discussed by Frazzini and Pedersen (2014) [2]. Lastly we introduce the propositions 1 and 2 from the same paper which we wish to test.

2.1 The Capital Asset Pricing Model

Our baseline model of interest is the capital asset pricing model (CAPM) which is a simple model that seeks to explain the relationship between expected returns and idiosyncratic risk (risk that cannot be diversified away). The model predicts this relationship to be linear and positive with the expected return of an asset being equal to the risk free rate plus a risk premium which is proportional to the asset's beta, which measures the sensitivity of the asset's returns to the returns of the market portfolio. The CAPM makes several underlying assumptions. This includes: Equilibrium model in a single-period economy, no transaction costs or taxes, risk-free borrowing and lending is universally available, investors share homogeneous expectations and investors maximize returns given limited risk.[4] The model can be specified as a regression like the following:

$$R_i = \alpha + \beta R_m + \epsilon \tag{2.1}$$

Where R_i is the return of asset i, α is the intercept, β is the slope coefficient which measures the sensitivity of the asset's returns to the returns of the market portfolio, R_m is the return of the market portfolio and ϵ is the error term. The CAPM predicts that in equilibrium, all assets should lie on the security market line (SML) which is a graphical representation of the relationship between expected returns and beta. The SML is upward sloping, indicating that higher beta assets should have higher expected returns.

Empirical evidence, however, has long challenged the predictions of the CAPM. One of the most well-known anomalies is the beta anomaly, which refers to the empirical finding that stocks with higher betas tend to have lower alphas than those with lower betas. This contradicts the predictions of the CAPM and suggests that there may be other factors at play in determining expected returns. The paper of focus here is the paper of Frazzini and Pedersen [2] (2014) which seeks to explain the beta anomaly and test the propositions that arise from their theoretical framework.

2.2 Leverage Constraints and a Flatter Security Market Line

The paper of Frazzini and Pedersen (2014)[2] suggests that the presence of leverage constraints can lead to a flatter security market line, which can explain the beta anomaly. Leverage constraints refer to the limitations that investors face when trying to borrow money to invest in higher beta assets. These constraints can arise from various sources, such as margin requirements, borrowing limits, or risk management policies. When investors are unable to borrow they compensate by over-buying high-beta risky assets which. This leads to excess demand which lower expected returns for the high beta assets and lowers their alpa's. This is the reason for the flatter security market line and the beta anomaly. The paper [2] provides a theoretical framework for understanding how leverage constraints can lead to a flatter security market line and tests this framework using empirical data from the US equity market.

2.3 Proposition 1: High Beta is Low Alpha

Frazzini and Pedersen (2014)[2] show that in the presence of funding constraints, the equilibrium expected return for any security s is:

$$E_t(r^s_{t+1}) = r^f + \psi_t + \beta^s_t \lambda_t \tag{2.2}$$

where $\lambda_t = E_t(r^M_{t+1}) - r^f - \psi_t$ is the risk premium adjusted for funding constraints and ψ_t is the average Lagrange multiplier measures the tightness of the funding constraints. A security's alpha with respect to the market is then:

$$\alpha^s_t = \psi_t(1 - \beta^s_t) \tag{2.3}$$

The alpha decreases in beta β^s_t. When $\beta^s_t > 1$, alpha is negative — high-beta stocks are overpriced. When $\beta^s_t < 1$, alpha is positive — low-beta stocks are underpriced. Furthermore, for an efficient portfolio, the Sharpe ratio is highest for a beta below one, decreasing in β^s_t for higher betas and increasing for lower betas.

2.4 Proposition 2: The BAB Factor

To construct the BAB factor, let w^L_t be the relative portfolio weights for a portfolio of low-beta assets with return $r^L_{t+1} = w^L_t r_{t+1}$, and similarly a portfolio of high-beta assets with return r^H_{t+1}. The betas of these portfolios are denoted β^L_t and β^H_t, where $\beta^L_t < \beta^H_t$. The BAB factor return is then:

$$r^{BAB}_{t+1} = \frac{1}{\beta^L_t}(r^L_{t+1} - r^f) - \frac{1}{\beta^H_t}(r^H_{t+1} - r^f) \tag{2.4}$$

This portfolio is market-neutral with a beta of zero. The long side is leveraged to a beta of one and the short side is de-leveraged to a beta of one. Frazzini and Pedersen (2014) [2] show that the expected excess return of this self-financing BAB factor is positive:

$$E_t(r^{BAB}_{t+1}) = \frac{\beta^H_t - \beta^L_t}{\beta^L_t \beta^H_t} \psi_t \geq 0 \tag{2.5}$$

The expected return is increasing in the ex ante beta spread $(\beta^H_t - \beta^L_t)/(\beta^L_t \beta^H_t)$ and funding tightness ψ_t. Thus a market-neutral BAB portfolio that is long leveraged low-beta securities and short higher-beta securities earns a positive expected return on average, with the size depending on how binding the funding constraints are in the market.

Chapter 3

Data and Methodology

In this chapter we describe the data and methodology used to test the propositions of Frazzini and Pedersen (2014) [2]. We first summarize the characteristics of the data and their sources, then we detail the beta estimation procedure, the construction of the BAB factor, and the Fama-MacBeth regressions used to test proposition 1 and 2. Finally, we discuss how we introduce and estimate transaction costs for the BAB strategy.

3.1 Data Description

This paper replicates the Betting Against Beta (BAB) factor of Frazzini and Pedersen [2] for the US equity market using CRSP data accessed through the `tidyfinance` package [3]. The sample covers January 1963 through December 2025. The choice of this period is driven by data availability and to ensure a long enough time series for robust beta estimation and portfolio construction.

Table 3.1: Data sources

Dataset	Frequency	Content	Purpose
CRSP stock data	Monthly	Excess returns, market capitalisation (NYSE, AMEX, NASDAQ)	BAB portfolio construction and factor returns
CRSP stock data	Daily	Excess returns, bid-ask prices	Volatility component of beta; proportional half-spreads for transaction costs
Fama-French factors	Monthly & daily	Risk-free rate, market excess return	Excess return computation; correlation component of beta

Note: CRSP data is accessed via WRDS through the `tidyfinance` Python package. Fama-French factors are downloaded from the Kenneth French data library. All data covers January 1963 to December 2025.

The WRDS credentials are stored by `tidyfinance` in its own configuration file. Run the one-time setup below before the first download. Leave this cell commented out once credentials are saved.

3.2 Fama-French Factors

We download Fama-French 3-factor data at monthly and daily frequency. We retain only the risk-free rate and market excess return, which are the inputs needed for excess return computation and beta estimation.

Table 3.2

	Mean	Std	P5	P25	Median	P75	P95
Excess Return	0.0093	0.1772	-0.2127	-0.0665	-0.0028	0.0676	0.2496
Beta	1.1706	0.5536	0.4733	0.8023	1.0782	1.4231	2.1433
Mkt Cap ($M)	3194.1899	31486.0677	4.1716	26.9820	130.1482	768.8673	9434.6907

3.3 Monthly CRSP Data

We download monthly CRSP data for all US common stocks listed on NYSE, AMEX, and NASDAQ. Market capitalisation is computed as shares outstanding times the absolute price, expressed in millions of USD. We also construct a one-month lagged market cap, which serves as the portfolio weight in the BAB factor construction.

3.4 Data Cleaning

Market capitalisation is computed as shares outstanding times absolute price, expressed in millions of USD. Observations with missing returns or market capitalisation are excluded from the sample. Excess returns are clipped at -100% to remove data errors in CRSP raw returns. Lagged market capitalisation (one month) is used as portfolio weights to ensure all weighting variables are known at the time of portfolio formation.

3.5 Daily CRSP Data

Beta estimation in Frazzini and Pedersen (2014)[2] requires daily returns. The volatility component of beta uses a 1-year rolling window (252 trading days, minimum 120 observations) of daily excess returns. We also retain bid and ask prices from CRSP to compute proportional half-spreads for the transaction cost analysis. We download the daily data in batches of 500 PERMNOs to manage memory.

3.6 Sample Description

Table Table 3.2 reports summary statistics for the key variables. Figure Figure 3.1 shows how many stocks are available each month, establishing the scope of the data, while Figure Figure 3.2 shows the cross-sectional distribution of estimated betas, which motivates the beta anomaly analysis that follows.

Note: Summary statistics for the final analysis sample covering January 1963 to December 2025. Excess Return is monthly and expressed as a decimal. Market Cap is in millions of USD. Beta is estimated following Frazzini and Pedersen (2014) as the product of a 60-month rolling correlation and a 252-day rolling volatility ratio, shrunk toward one. Source: CRSP via WRDS and Kenneth French data library.

3.7 Beta Estimation

Following Frazzini and Pedersen [2], we estimate beta as the product of a correlation component and a volatility component:

$$\hat{\beta}_i = \hat{\rho}_i \cdot \frac{\hat{\sigma}_i}{\hat{\sigma}_m}$$

The correlation $\hat{\rho}_i$ is estimated from a 5-year rolling window of monthly excess returns (minimum 3 years), while the volatility ratio is estimated from a 1-year rolling window of daily excess returns (minimum 120 observations). This two-component approach improves precision by using higher-frequency data for the noisier volatility estimate.

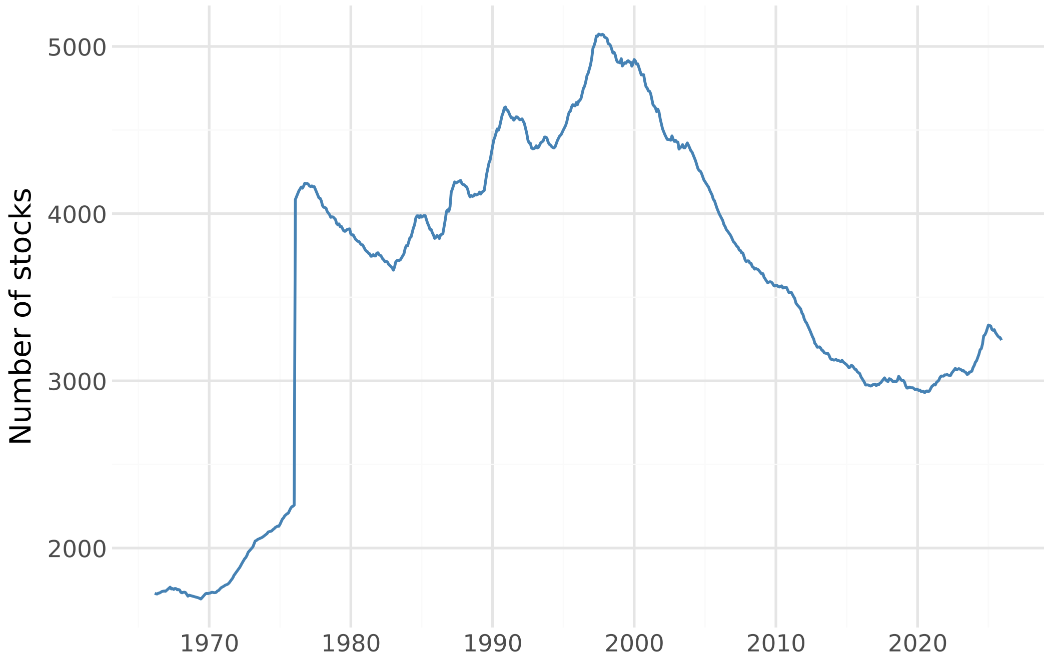

Figure 3.1: Number of distinct stocks included in the sample each month. Coverage grows steadily from around 500 stocks in the early 1960s to a peak of over 4,000 in the late 1990s before declining as NASDAQ listings contracted following the dot-com bubble. The pattern reflects the expansion of US equity markets and the availability of valid beta estimates.

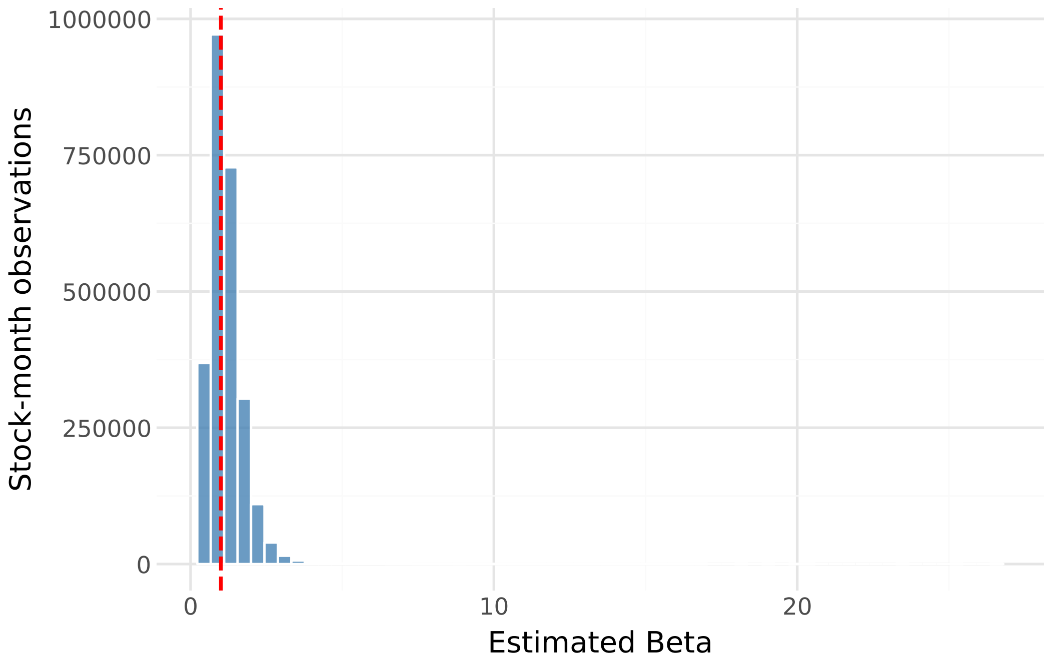

Figure 3.2: Cross-sectional distribution of estimated betas across all stock-month observations in the sample. The distribution is centred slightly below one with a long right tail driven by high-volatility small-cap stocks. The red dashed line marks a beta of one, which corresponds to the market portfolio under the CAPM.

3.7.1 Volatility Component

We estimate annualised standard deviations using a 252-day rolling window of daily excess returns based on yearly business days. The stock's standard deviation $\hat{\sigma}_i$ and the market standard deviation $\hat{\sigma}_m$ are estimated over the same window to ensure comparability. For each stock-month, we retain the last available daily estimate within the month.

3.7.2 Correlation Component

We estimate the correlation with the market $\hat{\rho}_i$ from a 60-month rolling window of monthly excess returns, requiring a minimum of 36 months of observations to obtain a valid estimate.

3.7.3 Combined Beta Estimate

The two components are combined into a time-series beta $\hat{\beta}_i^{ts} = \hat{\rho}_i \cdot (\hat{\sigma}_i/\hat{\sigma}_m)$, which is then shrunk toward the cross-sectional mean of one using Vasicek (1973) shrinkage with weights $(0.6, 0.4)$ as in Frazzini and Pedersen [2]:

$$\hat{\beta}_i = 0.6 \cdot \hat{\beta}_i^{ts} + 0.4 \cdot 1$$

Beta estimates are lagged by one month before merging with returns to avoid a look-ahead bias in portfolio construction.

3.8 Proposition 1: The Security Market Line

Proposition 1 predicts that the empirical SML is flatter than CAPM implies: high-beta assets earn lower risk-adjusted returns (negative CAPM alpha) and low-beta assets earn higher risk-adjusted returns (positive CAPM alpha). We test this by running Fama-MacBeth cross-sectional regressions of excess returns on lagged betas.

3.8.1 Beta-Sorted Portfolio Alphas

We sort stocks into 10 beta-decile portfolios and compute each decile's monthly CAPM alpha to visualise the alpha-beta relationship.

3.9 Proposition 2: The BAB Factor

Following Frazzini and Pedersen [2], at each month t we rank stocks by their estimated beta. We form a low-beta portfolio (below-median beta) and a high-beta portfolio (above-median beta). Portfolio weights are rank-based and sum to one within each leg. The BAB factor return is:

$$r_t^{BAB} = \frac{1}{\beta_t^L} r_t^L - \frac{1}{\beta_t^H} r_t^H$$

where β_t^L and β_t^H are the value-weighted average betas of the low and high portfolios.

3.10 Transaction Costs

We estimate the round-trip transaction cost for the BAB strategy using daily bid-ask spreads from CRSP. The proportional half-spread on day d for stock i is

$$c_{i,d} = \frac{\text{ask}_{i,d} - \text{bid}_{i,d}}{2 \cdot \text{mid}_{i,d}}$$

where $\text{mid}_{i,d}$ is the bid-ask midpoint. The monthly spread for each stock is the median of its daily half-spreads within the month, which is robust to days with stale or missing bid/ask quotes. We apply this cost to BAB portfolio turnover: each rebalancing incurs one half-spread on the sold leg and one on the bought leg. Following Frazzini, Israel, and Moskowitz (2015), we assume full monthly rebalancing, so this is an upper bound on realised costs.

Now we use this to test propositions 1 and 2 with and without transaction costs to see the effect of transaction costs on the performance of the BAB strategy.

Chapter 4

Results

In this chapter, we present the main results of our analysis. First we showcase the prevalence of the beta anomaly by testing proposition 1 and then we present the performance of the BAB-portfolio to test proposition 2. Lastly, we present the results of our extension of proposition 2 by introducing transaction costs into our analysis.

4.1 Proposition 1: The Security Market Line

To test proposition 1 we run Fama-MacBeth cross-sectional regressions of monthly excess returns on lagged betas. The results are presented below. The intercept is positive and highly statistically significant with a mean of 0.79% per month and a t-statistic of 5.84, indicating that there is a substantial return that is unrelated to market beta. More central to proposition 1, the slope coefficient on beta is 0.14% per month with a t-statistic of only 0.70, which is statistically indistinguishable from zero. Under the CAPM, this slope should equal the market excess return, which over our sample averages considerably higher. The near-zero and insignificant beta coefficient therefore indicates that the cross-sectional compensation for bearing systematic risk is far smaller than the CAPM predicts, consistent with a flatter-than-predicted security market line and with the beta anomaly. We note that the empirical SML plotted below appears to slope slightly downward, while the Fama-MacBeth coefficient is slightly positive. This apparent tension reflects different weighting: Fama-MacBeth weights all individual stocks equally within each month, so the moderate-beta stocks that constitute the bulk of the cross-section determine the average slope. The decile-based SML, by contrast, fits a line through ten equally-weighted group averages, giving the extreme high-beta decile (average beta 2.1) disproportionate leverage on the fitted slope. Both results are fully consistent with the beta anomaly and proposition 1: the empirical compensation for beta is economically negligible and statistically indistinguishable from zero, far below the CAPM prediction.

4.1.1 Beta-Sorted Portfolio Alphas

The Fama-MacBeth results are complemented by an analysis of CAPM alphas across beta-sorted decile portfolios. Sorting all stocks into ten groups by their estimated beta and computing each decile's average monthly CAPM alpha reveals a striking monotone pattern. Low-beta deciles earn positive and statistically significant alphas: decile 1 (average beta of 0.51) earns 0.27% per month, while deciles 2 and 3 earn 0.22% and 0.25% per month, both significant at conventional levels with t-statistics above 2.4 and 3.2, respectively. Moving up the beta distribution, alphas decline steadily. Decile 6 is

Table 4.1

	mean	t_stat
intercept	0.007863	5.836262
beta_coef	0.001418	0.703093

Table 4.2

	beta_decile	alpha	alpha_t	avg_beta
0	1	0.002741	1.746439	0.511693
1	2	0.002154	2.450884	0.676983
2	3	0.002496	3.205053	0.804396
3	4	0.001591	2.259553	0.916394
4	5	0.001141	1.801973	1.024017
5	6	-0.000145	-0.198470	1.135270
6	7	0.000748	0.748097	1.260848
7	8	-0.001768	-1.290402	1.418103
8	9	-0.004623	-2.633372	1.641384
9	10	-0.008241	-3.164249	2.103706

essentially zero ($-0.01=$−0.20), and the three highest-beta deciles earn negative and increasingly significant alphas. Decile 9 (average beta 1.64) earns $-0.46=$−2.63), and the top decile (average beta 2.10) earns $-0.82=$−3.16). This monotone decline in alpha across the beta distribution is precisely what proposition 1 predicts. The two panels below visualise this result: the upper panel plots the empirical security market line (solid blue) against the CAPM prediction (red dashed), showing that the empirical line is clearly flatter; the lower panel displays the CAPM alpha for each decile, illustrating the transition from positive alphas at the low end to large negative alphas at the high end.

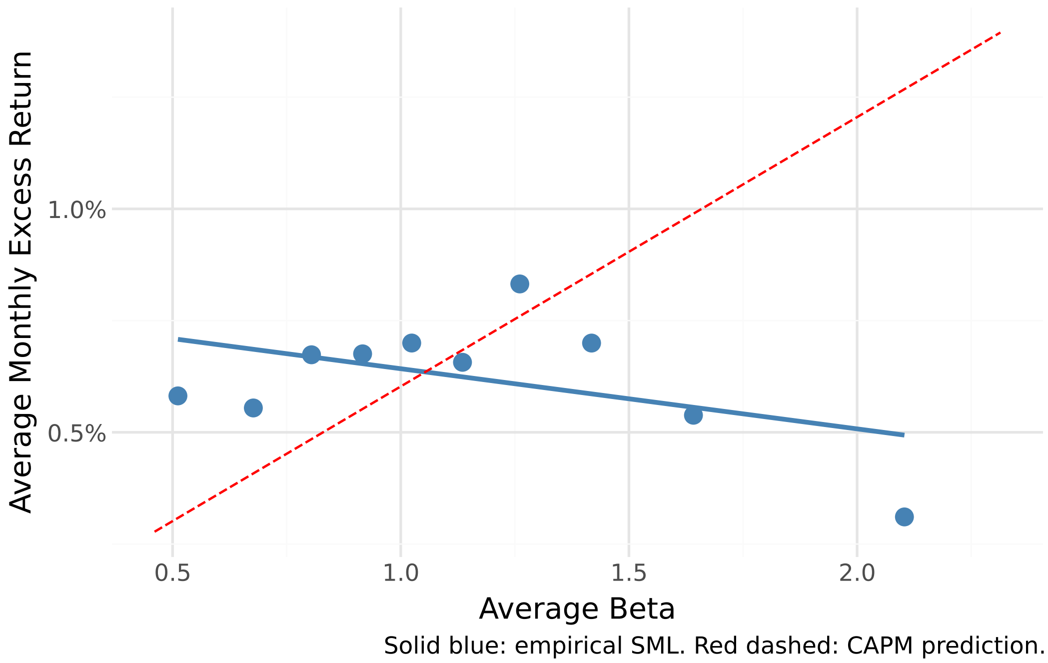

Figure 4.1

4.2 Proposition 2: The BAB Factor

Having established the presence of the beta anomaly, we now turn to proposition 2 and examine whether a BAB portfolio that exploits this anomaly generates positive excess returns. Following Frazzini and Pedersen (2014), we construct the BAB factor each month by going long a leveraged portfolio of below-median-beta stocks and short a de-leveraged portfolio of above-median-beta stocks, with both legs scaled to a beta of one. The performance statistics are reported below. The BAB factor earns an annualised mean return of 6.72% with an annualised standard deviation of 25.66%, yielding

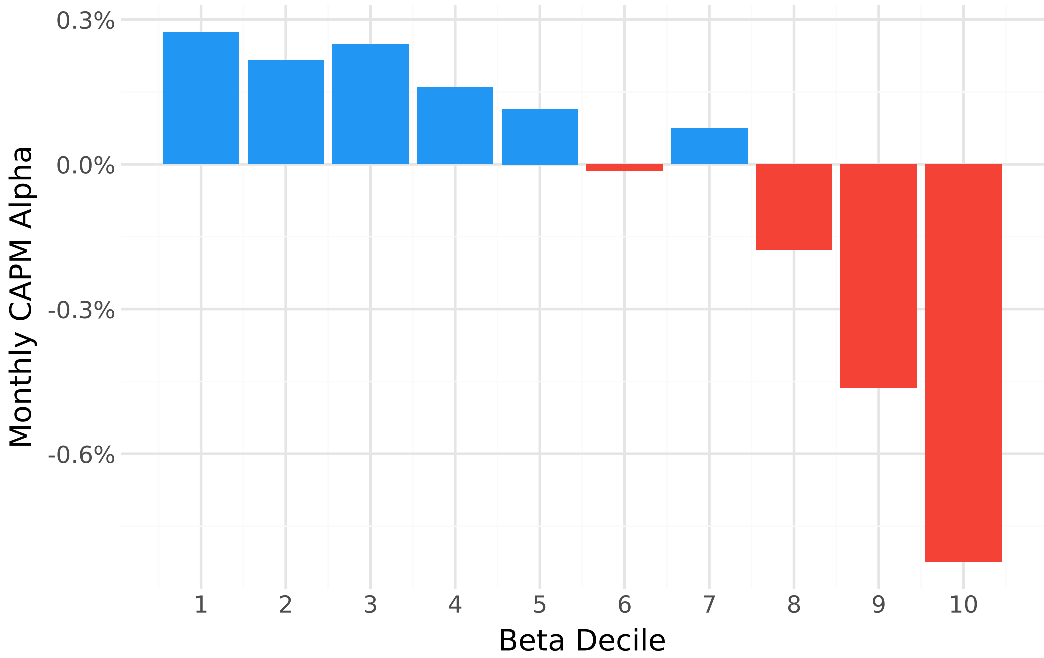

Figure 4.2

Table 4.3

	BAB Factor
Ann. Mean Return	0.0672
Ann. Std Dev	0.2566
Sharpe Ratio	0.2620
t-statistic	2.0267

a Sharpe ratio of 0.262. The associated t-statistic of 2.03 indicates that the positive mean return is statistically significant at the 5% level, providing support for proposition 2. The cumulative return chart below illustrates the time-series behaviour of the strategy: starting from $1 in the early 1960s, the portfolio grows to a peak of approximately $7.50 around the turn of the millennium before declining, and recovers to a similar level by the end of the sample in the early 2020s. The strategy thus generates meaningful returns over the full sample period, albeit with considerable variation across sub-periods.

4.3 Transaction Costs

While the gross BAB factor earns positive returns, a real-world investor must also bear the transaction costs of monthly rebalancing. To assess this, we estimate round-trip costs from daily CRSP bid-ask spreads and apply them to the turnover of each portfolio leg. The results, reported below, reveal a dramatic impact of trading frictions. Over the subsample for which transaction cost data are available, the gross BAB strategy earns an annualised mean return of 5.56% (t$=$1.07), but after deducting estimated transaction costs the net return falls to $-161.6=$−18.14). The average annual transaction cost drag amounts to 167.1 percentage points, far exceeding the gross return. This implies that, under the assumption of full monthly rebalancing, the BAB strategy is entirely non-viable after accounting for realistic trading frictions. The result should be interpreted as an upper bound on actual transaction costs, since in practice investors would rebalance less frequently and only trade when the cost savings exceed the friction. Nonetheless, it highlights that the gross profitability documented in section 4.2 is highly sensitive to implementation costs, particularly given the wide bid-ask spreads prevalent in CRSP data for earlier decades prior to market decimalization.

Figure 4.3

Table 4.4

	Gross BAB	Net BAB (after TC)	Avg annual TC drag
Ann. mean return	0.0556	-1.6157	1.6714
t-statistic	1.0664	-18.1386	NaN

Chapter 5

Discussion

In this chapter, we discuss the implications of our findings and how they relate to existing literature on the beta anomaly and the performance of the BAB-portfolio. We also discuss the limitations of our analysis and suggest avenues for future research.

relate to main paper.. suggest further improvements such as transaction costs(bid ask spread), risk-adjusted returns, and other markets.

explore shortcomings of data and methodology such as the use of historical data, the choice of time period, and the potential for survivorship bias.

Chapter 6

Conclusion

This paper contributed to the research of market anomalies by testing the propositions 1 and 2 from the paper of Frazzini and Pedersen (2014) [2] which seek to explain the beta anomaly and the performance of the BAB strategy. Using data from US equity markets we find evidence of a negative relationship between beta and alpha, which supports proposition 1 and that the beta anomaly is present leading to a flatter market security line. We also find that the BAB-portfolio generates positive returns, which supports proposition 2. However, when we introduce transaction costs into our analysis, we find that the performance of the BAB-portfolio is significantly reduced, suggesting that the strategy may not be as effective as initially thought when accounting for real-world trading frictions. The limitations of this paper include the use of a simple measure of transaction costs and the focus on US equity markets, which may limit the generalizability of our findings. Future research could explore more sophisticated measures of transaction costs, such as market impact, and examine the performance of the BAB strategy in other markets and asset classes. Additionally, future research could investigate the role of other factors, such as liquidity and investor sentiment, in explaining the beta anomaly and the performance of the BAB strategy.

Chapter 7

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